1 **Design & Implementation of Human-Computer Interfaces**
NPTEL-MOOCS

Dr Samit Bhattacharya
Computer Science and Engineering
IIT Guwahati

2 **L6 – Contextual Inquiry**

Dr Samit Bhattacharya
Computer Science and Engineering
IIT Guwahati

3 **Recap**

- We are discussing the requirement gathering stage of ISDLC
  - Learned about the need for usability requirement gathering – a type of non-functional requirements

4

5 **Recap**

- In this lecture – shall learn about one usability requirement gathering technique: contextual inquiry

6 **Motto**

- Know thy user

7 **Recap - Usability Definition**

- ISO definition (ISO 9241-210:2009) - "the extent to which a product can be used by specified users to achieve specified goals with effectiveness, efficiency, and satisfaction in a specified context of use."

8 **Contextual Inquiry**

- A method to gather end-user requirements
  - Same as usability requirements
  - One of many such methods

9 **Contextual Inquiry**

- Its importance in UCD highlighted as early as 1988, although the term used was "phenomenological research method"
- By the end of the 1990s, it became well developed and popular

10 **Contextual Inquiry**

- Often referred to as a "semi-structured" interview process
  - Consists of both interview and observations

11 **Contextual Inquiry**

- Primary objective - watch and observe users perform tasks in their "natural" work setting (the work context)
- During observations, users may be asked questions to clarify on certain behavior or activities

12 **Contextual Inquiry**

- Usually the process continues for one to two hours

13 **Contextual Inquiry**

- Observations (including interaction with users) enable designers learn about task domain, work culture and physical and social constraints at the work place
  - The knowledge is important since the "specific work context" is an essential part of the definition of usability

14 **Contextual Inquiry**

- Ex - recollect the calendar app example from the last lecture

15 **Contextual Inquiry**

- Suppose the target group of users are the managerial staff of any organization

16 **Contextual Inquiry**

- Without any knowledge about intended user behavior, we can come up with all sorts of possible designs

17 **Contextual Inquiry**

- However, suppose we now performed a CI and observed two important activities performed by the users: set meeting dates and reminders

18 **Contextual Inquiry**

- Clearly, a calendar app intended to be used by this group of users should have support for these activities

19 **CI - Types**

20 **Two Ways**

- The inquiry can be performed in either way
  - Active mode
  - Passive mode

21 **Active Mode**

- Observer is physically present

22 **Passive Mode**

- User activities recorded/videotaped and reviewed later

23 **Contextual Inquiry**

- During the inquiry, observer should act as "apprentice" and treat the users as "masters"
- Purpose is to learn user behavior, not to teach them how to behave

24 **CI - Stages**

25 **Five Stages**

- Plan
- Initiate
- Execute
- Close
- Reflect

26 **Plan (Things to do)**

- Identification of the goals of observation
- Knowledge gathering about task domain
- Arrangements for recording observations
- Identification of users and the date, time and place of observation
- Scripting of the procedure and rehearsals

27 **Initiate Stage**

- Contact the authorities for permission and open communication with the end users to remove their anxieties

28 **Execute Stage**

- Perform observations and record data

29 **Close Stage**

- Send "thank you" notes to the users (you may require them later)

30 **Reflect Stage**

- Analyze the data collected to identify design goals
- Not Easy since you may end up with lots of data
- Tools and techniques help - e.g, affinity diagram method

31 **Affinity Diagram & Use**

32 **Affinity Diagram Method**

- FIVE steps
  1. Generate Idea
  2. Display ideas
  3. Sort ideas into groups
  4. Create "group header" (names)
  5. Draw finished diagram

33 **Example**

- Creation of affinity diagram of a Class-note Taking Application, focusing mainly on the needs of college-going students

34 **Example**

- Assume CI done and usability requirements identified!
- 3 tools used to study user behavior and gather requirements

- Physical note taking (pen & paper based)
- Notion.so web application
- xournal++ desktop software

35 **Example**

- Observations noted down and analyzed (following stages of affinity diagram method)

36 **Step 1 & 2 – Generate & Display Ideas**

37 **Step 1 & 2 – Generate & Display Ideas**

38 **Step 3 – Sort into Groups**

39 **Step 4 – Header**

40 **Step 5 –Final Drawing**

41 **Reflection**

- Can come up with a set of design recommendations such as

"For correcting mistakes, there should be a provision to erase content. This would help in keeping the notes neat and tidy"

- Note: we are not talking about "how to do that" but "what to do to make it usable"

42 **Book**

- Bhattacharya, S. (July, 2019). Human-Computer Interaction: User-Centric Computing for Design, McGraw-Hill India

• Print Edition: ISBN-13: 978-93-5316-804-9; ISBN-10: 93-5316-804-X
• E-book Edition: ISBN-13: 978-93-5316-805-6; ISBN-10: 93-5316-805-8

Chapter 2, Sec 2.4.3